

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Chemistry — SSC-I (9th Grade)

Syllabus: Chapter 11 — Environmental Chemistry: Air

Total Marks: 65

Time Allowed: 2h 30m

Version: 1

General Instructions:

- Write your name, roll number and section clearly on the answer sheet.
- Section A is compulsory. Encircle the correct option on the question paper itself.
- In Section B, attempt **all 11** questions. For each, attempt either the main question **OR** its alternative.
- In Section C, attempt **all 4** questions. For each, attempt either the main question **OR** its alternative.
- Use black or blue ink only. Pencil is not allowed except for diagrams.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	The approximate percentage of nitrogen in clean, dry air is:	21%	50%	78%	0.93%	(A)(B)(C)(D)
2	Sulphur dioxide is produced mainly from:	vehicle exhaust only	burning of fossil fuels containing sulphur	decomposition of vegetation	photosynthesis	(A)(B)(C)(D)
3	Carbon monoxide is dangerous because it:	has a strong smell	combines with haemoglobin to form carboxyhaemoglobin	reacts with ozone	causes acid rain	(A)(B)(C)(D)
4	NO ₂ is described as a:	colourless, odourless gas	reddish-brown gas with pungent odour	blue gas with sweet smell	colourless gas with irritating odour	(A)(B)(C)(D)
5	Greenhouse gases act like a blanket by:	reflecting all sunlight back to space	absorbing and re-radiating infrared radiation	producing oxygen in the atmosphere	filtering ultraviolet radiation	(A)(B)(C)(D)
6	Acid rain is defined as rain with a pH:	greater than 7	equal to 7	less than 5.6	between 6 and 7	(A)(B)(C)(D)
7	Ground-level ozone is classified as a:	primary pollutant from car exhausts	secondary pollutant from NO _x + VOCs in sunlight	beneficial gas protecting against UV	greenhouse gas causing global warming	(A)(B)(C)(D)
8	The catalysts used in catalytic converters include:	iron and nickel	platinum, palladium, and rhodium	calcium oxide and copper	sodium and potassium	(A)(B)(C)(D)
9	Which gas is primarily responsible for both acid rain AND global warming?	Ozone	SO ₂	CO ₂	N ₂	(A)(B)(C)(D)
10	Methane enters the atmosphere from livestock mainly through:	blood circulation	photosynthesis	belching and flatulence during digestion	burning of fossil fuels	(A)(B)(C)(D)
11	Which reaction correctly shows SO ₃ forming sulphuric acid?	SO ₃ + H ₂ → H ₂ SO ₄	SO ₃ + H ₂ O → H ₂ SO ₄	SO ₃ + O ₂ → H ₂ SO ₄	SO ₃ + 2H ₂ O → H ₂ SO ₃	(A)(B)(C)(D)
12	Photosynthesis removes which pollutant from the atmosphere?	SO ₂	NO ₂	CO ₂	CO	(A)(B)(C)(D)

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt all 11 questions. For each pair, attempt either the main question OR its alternative.

Part	Question	Marks	OR	OR Question	Marks
2(i)	State the composition of clean, dry air by percentage volume. Name the two most abundant gases and give the percentage of each.	3	OR	Define atmosphere. What is water vapour? Why does the amount of water vapour in air vary from place to place?	3
2(ii)	Define pollutant. List any four major air pollutants and state one harmful effect of each.	3	OR	Differentiate between natural sources and human activity sources of air pollution. Give two examples of each.	3
2(iii)	Describe the physical properties and harmful effects of sulphur dioxide (SO ₂) on human health.	3	OR	Write the balanced equations for: (a) C + O ₂ (limited) → CO (b) S + O ₂ → SO ₂ (c) N ₂ + O ₂ → 2NO.	3
2(iv)	What is carbon monoxide? Why is it called a colourless, odourless, poisonous gas? Explain how it causes death.	3	OR	What are particulates? Name three examples and explain their harmful effects on human health.	3
2(v)	Define greenhouse effect. Name three greenhouse gases and explain how they trap heat in the atmosphere.	3	OR	Define global warming. List four consequences of global warming if it continues unchecked.	3

2(vi)	What is methane? State three sources of methane in the atmosphere. Why does methane contribute to global warming?	3	OR	What is ground-level ozone? How is it formed? State two harmful effects on humans and the environment.	3
2(vii)	Define acid rain. State the pH of normal rainwater and of acid rain. Name the two acids most responsible for acid rain.	3	OR	Write the sequence of reactions that converts SO ₂ into sulphuric acid in the atmosphere. Include all three equations.	3
2(viii)	Describe two effects of acid rain on: (a) marble structures and (b) aquatic life. Include the balanced equation for marble damage.	3	OR	How does acid rain affect metals? Write the balanced equation for iron reacting with sulphuric acid. Why is acid rain more dangerous than normal rain?	3
2(ix)	What is a catalytic converter? Name the catalysts it contains. Write the equation for the conversion of NO and CO into harmless products.	3	OR	State three strategies to reduce acid rain. Explain how flue gas desulphurization works.	3
2(x)	Define photosynthesis. Write the balanced equation and explain its role in reducing CO ₂ in the atmosphere.	3	OR	State three strategies to reduce climate change (global warming). For each, briefly explain how it helps.	3
2(xi)	What are personal protective measures against air pollution? Describe the use of (a) N95 masks and (b) carbon monoxide detectors.	3	OR	What is an air quality index? How can it help reduce exposure to air pollutants? Give two practical examples.	3

SECTION C — Long Questions (4 × 5 = 20 Marks) — Attempt all 4

Part	Question	Marks	OR	OR Question	Marks
3(i)	(a) Describe the six major air pollutants: SO _x , NO _x , CO, CO ₂ , CH ₄ , and ground-level ozone. For each, state its main source and one harmful effect. [4] (b) Distinguish between primary and secondary pollutants. Give one example of each. [1]	5	OR	(a) Describe natural and human sources of air pollution. Name three natural processes and four human activities that release air pollutants. [3] (b) Write balanced equations for: (i) incomplete combustion of carbon (ii) S + O ₂ → SO ₂ (iii) N ₂ + O ₂ → 2NO (iv) 2NO + O ₂ → 2NO ₂ . [2]	5
3(ii)	(a) Explain the greenhouse effect in detail. Name the greenhouse gases; describe how infrared radiation is emitted by Earth's surface and how greenhouse gases trap this energy. [3] (b) Define global warming. Explain how human activities have intensified the greenhouse effect. List three consequences of continued global warming. [2]	5	OR	(a) Describe strategies to reduce climate change under: (i) transition to renewable energy (ii) afforestation and reforestation (iii) reducing CO ₂ from transport. [3] (b) Explain the role of the Paris Agreement. State its two key temperature targets. [2]	5
3(iii)	(a) Define acid rain and explain its formation. Write the full sequence of equations showing how SO ₂ is converted to sulphuric acid and how NO ₂ is converted to nitric acid. [3] (b) Describe four harmful effects of acid rain on: (i) marble (ii) iron metals (iii) aquatic life (iv) trees. Include one balanced equation. [2]	5	OR	(a) Describe strategies to solve acid rain: (i) emission control (ii) flue gas desulphurization (iii) low sulphur fuels. [3] (b) Explain how catalytic converters reduce acid rain precursors. Write the conversion equation and state the catalysts used. [2]	5
3(iv)	(a) Define photosynthesis. Write the balanced equation. Explain how it removes CO ₂ and maintains the carbon cycle. [3] (b) List and explain three risk factors that cause long-term respiratory problems from air pollution. Name one disease linked to each. [2]	5	OR	(a) Explain how catalytic converters control air pollution from car exhausts. Name the catalysts, write the main equation, and state what harmful gases are converted. [3] (b) Describe three personal protective measures against air pollutants. For each, explain how it works. [2]	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Chapter 11 — Environmental Chemistry: Air (pp. 151-162)

— End of Question Paper —